Apex Precision Manufacturing Pvt. Ltd.

CNC Machine Operation Manual

CONTROLLED SAMPLE DOCUMENT FOR RAG CHATBOT TESTING

Document ID	APM-PROD-MAN-001	**Department**	Production
Version	1.0 – Sample	**Effective date**	30 October 2026
Document owner	Production Department	**Review status**	Training sample
Classification	Internal – Fictional	**Approval**	Not approved for real operations

IMPORTANT NOTICE

This document is fictional sample content created solely to develop and test a Retrieval-Augmented Generation (RAG) chatbot. It is not an approved company procedure, machine-specific instruction, legal policy, or substitute for manufacturer documentation, site risk assessments, qualified supervision, or applicable laws. Do not use it to operate, repair, inspect, or service real equipment.

Purpose

Provide a controlled training reference describing the general workflow for preparing, operating, and shutting down a CNC machining center.

Document control

The document owner is responsible for review and controlled distribution in a real organization. For this training dataset, version 1.0 is fixed and all names, policies, workflows, and examples are fictional.

1. Scope and responsibilities

• Applies to trained production personnel working with CNC machining centers at the Apex Precision Manufacturing training facility.

• Only employees who have completed machine-specific training and have written authorization may operate equipment.

• The machine manufacturer's manual and site-approved machine-specific risk assessment always take precedence over this generic sample document.

2. Before operation – readiness checks

• Confirm the work order, part drawing, revision, material, and approved setup sheet match the job.

• Check that the work area is clean, lighting is adequate, and access routes are clear.

• Inspect visible guards, interlocks, emergency-stop devices, and warning indicators without bypassing or defeating any safety device.

• Verify that the correct tooling, workholding, and measuring equipment are available and identified for the job.

• Check that required PPE is available and worn as specified by the site risk assessment.

3. Setup and program verification

• A qualified setter must verify the approved CNC program number and revision against the job traveler.

• Confirm tool identification and offsets using the approved setup sheet and machine-specific procedure.

• Secure the workpiece using the approved workholding method; do not improvise clamps or fixtures.

• Complete an independent setup check where required by the work order or local procedure.

• Keep doors and guards closed during automatic operation.

4. Production monitoring

• Run the first-off inspection using the current drawing and inspection plan before continuing batch production.

• Monitor alarms, unusual vibration, sound, temperature indications, and visible coolant or lubrication issues.

• Stop the cycle and notify the supervisor if a safety device, guard, interlock, or machine function appears abnormal.

• Record production quantity, scrap, downtime, and inspection results in the approved production record.

5. Shutdown and handover

• End the cycle using the machine-specific shutdown sequence supplied by the manufacturer and site engineering team.

• Leave the machine in the designated safe condition; remove chips only with approved tools and after hazardous motion has stopped.

• Report faults, incomplete work, tooling concerns, and abnormal observations to the next shift.

• Complete the job traveler and housekeeping checklist.

6. Records

• Setup verification checklist; first-off inspection record; production log; downtime and fault report; operator training record.

Revision history

Version	Date	Description	Prepared by
1.0	30-Oct-2026	Initial fictional sample for RAG testing	Project team